CLAIMS

What is claimed is:

1. A memory interface system for a computer, comprising:

- a memory slot on a motherboard, said slot containing a first two-dimensional array of individually addressable electromagnetic radiation emitters; and
- a removable memory module, said module containing memory chips and a second two-dimensional array of electromagnetic radiation sensors, wherein said second array is positioned to optically couple with the first array when the module is installed in the slot.

2. The system of claim 1, wherein data is transmitted between the motherboard and the memory module as a sequence of frames, with each frame comprising a plurality of data packets encoded as the emission characteristics of the emitters in the first array.

3. The system of claim 2, wherein the emission characteristics include at least one of wavelength or intensity.

4. The system of claim 1, wherein the emitters are micro-LEDs or micro-OLEDs and the sensors are a CMOS image sensor.

5. The system of claim 1, wherein the aggregate data transmission bandwidth of the interface is in the terabit-per-second or petabit-per-second range.

6. A memory module for a computer system, comprising:

- a plurality of memory integrated circuits (chips);
- a printed circuit board on which the memory chips are mounted; and
- a two-dimensional array of photon sensors integrated onto the module, configured to serve as the primary data interface for the module.

7. The memory module of claim 6, wherein the array of photon sensors replaces a traditional metallic edge connector.

8. A method for accessing computer memory, comprising the steps of:

- generating, by a memory controller, a sequence of data frames, wherein data to be written to a memory module is encoded into the emission characteristics of a two-dimensional array of emitters located in a motherboard memory slot;
- transmitting said sequence of frames as light from the emitter array;
- capturing, by a sensor array on the memory module, said sequence of frames; and
- decoding said frames and writing the data to memory chips on the module.

9. The method of claim 8, wherein the data is transferred at a rate exceeding 1 terabit per second.

10. The method of claim 8, further comprising a self-calibration step upon installation of the module, wherein the memory module and the memory slot electronically negotiate a data map to optimize accuracy.